

SCBG Guardian: A Structure-Constrained Runtime Safety Layer for Behavior Stabilization

We present SCBG Guardian, a runtime safety layer designed to stabilize behavior in dynamical systems by leveraging structure-constrained mode transitions. Unlike conventional safety mechanisms based on threshold-triggered rules, SCBG Guardian introduces a structural perspective in which behavior is regulated through role-semantic invariants and dynamic mode selection.

The system operates as an auxiliary layer that monitors system states and intervenes only when instability or unsafe conditions are detected. Instead of enforcing hard constraints, SCBG Guardian enables transitions between behavioral modes such as approach, avoidance, and stabilization.

Through a minimal simulation study, we compare SCBG Guardian against both a nominal controller and a threshold-based safety mechanism. The results show that SCBG Guardian successfully prevents collisions while maintaining task progression, whereas threshold-based approaches either fail to avoid collision or lead to stalled behavior.

These findings suggest that behavioral safety can be reframed as a problem of structure-driven mode selection, rather than constraint enforcement.

1. Introduction

Ensuring safety and stability in autonomous systems remains a fundamental challenge. Existing approaches typically rely on: • explicit safety constraints • threshold-based switching rules • optimization-based control

However, these approaches often exhibit limitations: • over-conservative behavior • inability to recover from instability • failure under conflicting objectives

This work proposes an alternative perspective:

Behavioral safety is not merely a constraint enforcement problem, but a structural mode transition problem.

We introduce SCBG Guardian, a runtime layer that stabilizes behavior by dynamically selecting modes based on structural invariants.

1. System Overview

SCBG Guardian is designed as a plug-in layer that operates alongside existing controllers.

Architecture

Nominal Controller ↓ Risk Monitor ↓ SCBG Guardian ↓ Final Control Output

Design Principles • Non-intrusive: does not replace existing controllers • Event-driven: activates only under risk • Reversible: returns control after stabilization

1. Core Mechanism

3.1 Structural Representation

SCBG Guardian is based on four structural components: • Approach (target-directed behavior) • Avoidance (obstacle-directed behavior) • Stabilization (damping and smoothing) • Switching (mode transition control)

3.2 Mode Selection

Behavior is generated as:

$$u = F(A, B, C, D)$$

where the dominance of each component is determined by structural relations: • danger level • conflict between approach and avoidance • system stability

3.3 Risk Detection

The system evaluates risk based on: • proximity to obstacle • oscillatory instability • trajectory inconsistency

When risk exceeds a threshold, control is transferred to the Guardian.

1. Experimental Setup

Environment • 2D point agent • Single target • Single obstacle

Controllers Compared

System Description	Original	Nominal controller	Threshold	Distance-based	safety	switching
SCBG Guardian	Structure-based	mode switching				

1. Results

5.1 Original Controller • Reaches target • Collides with obstacle • Fails task

5.2 Threshold-Based Safety • Avoids collision • Often stalls or deviates excessively • Fails to complete task

5.3 SCBG Guardian • Avoids collision • Maintains forward progress • Successfully completes task

5.4 Key Observation

SCBG Guardian differs fundamentally from threshold-based approaches: • Threshold → rigid switching • SCBG → dynamic mode blending

1. Discussion

6.1 Not a Replacement Controller

SCBG Guardian does not aim to optimize performance, but to ensure: • safety • recoverability • continuity

6.2 Structural Interpretation

The system demonstrates that:

Safety emerges from proper mode transitions rather than fixed constraints.

6.3 Generalization Potential

The approach may extend to: • robotics • autonomous driving • multi-agent systems

1. Conclusion

We introduced SCBG Guardian as a structure-driven runtime safety layer.

The results indicate that: • behavior stabilization can be achieved through structural mode transitions • safety can be reframed as a mode selection problem

SCBG Guardian provides a new perspective on behavioral safety by shifting from constraint enforcement to structure-driven mode selection.

SCBG Guardian 证明了：安全不是靠“限制”，而是靠“切换到正确的行为模态”